

PRIMPRI CHINCHWAD EDUCATION TRUST'S
PIMPRI CHINCHWAD COLLEGE OF ENGINEERING



Department of AS&H
Report of FA-2 Activity

Subject: Statistical Data Analysis Using R

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Date: 15 Oct. 2025

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Project Title: Movie Data Analysis Based on Movie Genres

Introduction:

The Movies Genre Dataset Analysis Project aims to explore and clean a large collection of movie-related data to gain meaningful insights into the global film industry. The dataset contains essential information about each movie, such as its IMDb ID, popularity score, budget, revenue, and original title.

Before performing any analysis, it is crucial to ensure that the dataset is clean, accurate, and free from duplicate records. Duplicate data entries can lead to misleading results and incorrect conclusions. Therefore, the first step of this project focuses on data cleaning using R, particularly on identifying and removing duplicate rows based on unique identifiers like the *IMDb ID* or *movie title*.

This project demonstrates the importance of data preprocessing in data science and highlights how clean data ensures reliable and insightful analysis.

Dataset Description:

This project is based on the Movies Genre Dataset, which provides detailed information about various movies collected from multiple sources such as IMDb and The Movie Database. The dataset contains essential attributes that describe different aspects of a movie from its financial performance to creative details and audience reception.

The dataset includes the following key columns:

1. **Id & imdb_id:**
 - **Datatype:** Numeric
 - **Description:** Unique identifiers for each movie entry.
2. **Popularity:**
 - **Datatype:** Numeric
 - **Description:** A numeric score indicating the movie's popularity.
3. **Budget and revenue:**
 - **Datatype:** Numeric
 - **Description:** Represent the production cost and total earnings of the movie.
4. **Original_title:**
 - **Datatype:** String
 - **Description:** The official name of the movie.

5. cast & director:

- **Datatype:** String
- **Description:** Provide information about the main actors and the person who directed the film.

6. Tagline, keywords, and overview:

- **Datatype:** String
- **Description:** Contain short descriptions and key phrases summarizing the movie's theme or plot.

7. Runtime:

- **Datatype:** Numeric
- **Description:** The total duration of the movie in minutes.

8. Genres:

- **Datatype:** Categorical (Action, Adventure, Animation, Comedy, Crime, Documentary, Drama, Family, Fantasy, Foreign, History, Horror, Music, Mystery, Romance, Science Fiction, TV Movie, Thriller, War, Western)
- **Description:** Specifies the type or category of the movie (e.g., Action, Drama, Comedy).

9. production:

- **Datatype:** String
- **Description:** Denotes the production company involved.

10. release_data and release_year:

- **Datatype:** Alpha-numeric
- **Description:** Indicate when the movie was released.

11. vote_count and vote_average:

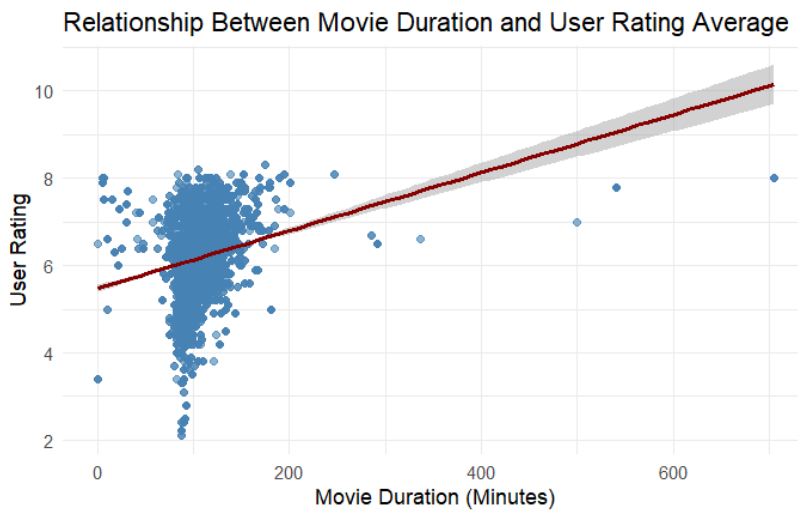
- **Datatype:** Numeric
- **Description:** Show the number of votes and the average rating received by the movie.

12. budget_adj and revenue_adj:

- **Datatype:** Numeric
- **Description:** Represent the adjusted budget and revenue values considering inflation or normalization factors.

Problem Statement 1: Is there a relationship between Movie Duration and User Rating?

The analysis shows that Adventure, Action, Fantasy, and Science Fiction movies generally have longer runtimes and higher ratings, while Western, Foreign, and Documentary genres tend to have shorter runtimes and lower ratings. Drama and Thriller movies are produced most frequently but receive average ratings overall.



R Script:

```
ggplot(movies, aes(x = runtime, y = vote_average)) + geom_point(alpha = 0.6, color =  
"steelblue") + geom_smooth(method = "lm", se = TRUE, color = "darkred") +  
  labs(title = "Relationship Between Movie Duration and User Rating Average", x = "Movie  
Duration (Minutes)", y = "User Rating") +  
  theme_minimal()
```

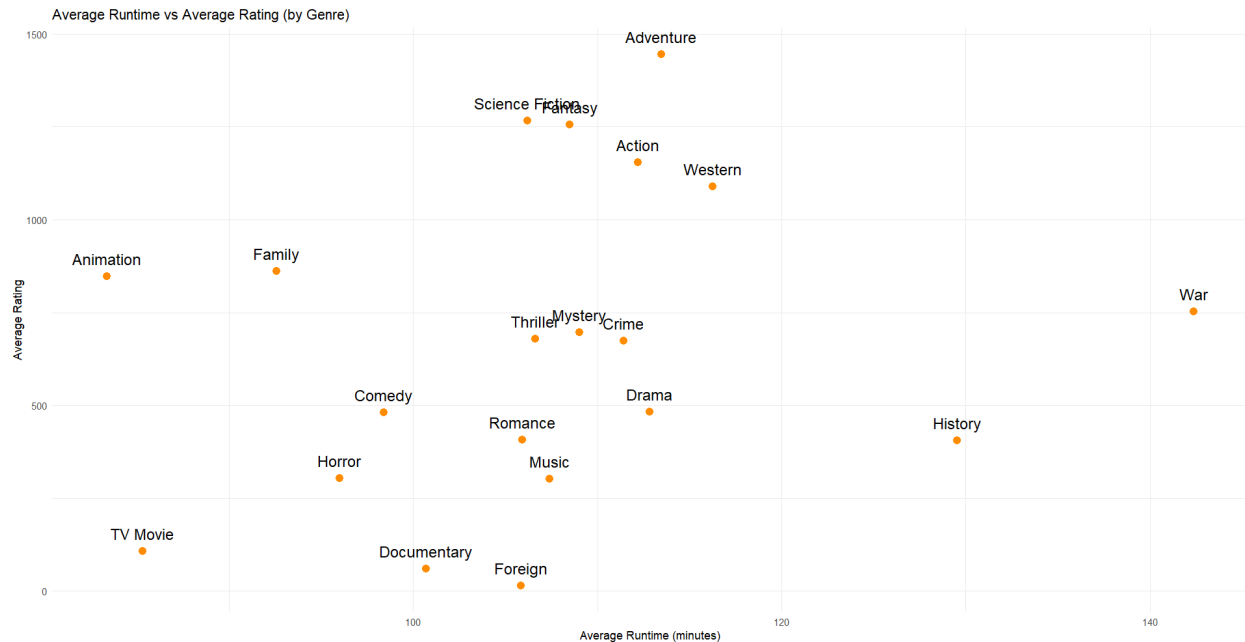
Observations:

The scatter plot shows a weak positive correlation between movie duration and user ratings. This indicates that longer movies generally receive slightly higher ratings, but the relationship is not strong, meaning other factors also influence ratings significantly.

Problem Statement 2: What does the analysis of average runtime by genre reveal about movie genres, their runtimes, and ratings? & What does the graph of average runtime vs average rating by genre reveal about the relationship between movie length and user ratings across different genres?

The problem statement analyzes how movie genres differ in average runtime, ratings, and production count. The R script groups movies by genre to find these averages. The analysis reveals that Adventure, Action, Fantasy, and Science Fiction films generally have longer runtimes and higher ratings, while Western, Foreign, and Documentary genres have shorter runtimes and lower ratings, offering insights into audience preferences and production trends.

	genres	avg_runtime	avg_rating	count
1	War	142.36364	752.61364	44
2	History	129.53571	405.71429	56
3	Western	116.25000	1090.15000	20
4	Adventure	113.46260	1446.10526	361
5	Drama	112.83294	483.57995	838
6	Action	112.20301	1153.48308	532
7	Crime	111.43511	674.11450	262
8	Mystery	109.02703	696.31757	148
9	Fantasy	108.48515	1256.22277	202
10	Music	107.38889	301.94444	72
11	Thriller	106.62783	678.29043	575
12	Science Fiction	106.21306	1266.99313	291
13	Romance	105.92784	407.26460	291
14	Foreign	105.83333	15.00000	6
15	Documentary	100.70968	59.37634	93
16	Comedy	98.38906	480.84062	640
17	Horror	96.01832	303.77656	273
18	Family	92.58369	860.90987	233
19	TV Movie	85.31250	107.43750	16
20	Animation	83.37059	847.95294	170



R Script:

```
avg_runtime_genre <- movies %>% group_by(genres) %>%
  summarise(
    avg_runtime = mean(runtime, na.rm = TRUE),
    avg_rating = mean(vote_count, na.rm = TRUE),
    count = n()
  ) %>% arrange(desc(avg_runtime))
View(avg_runtime_genre)
```

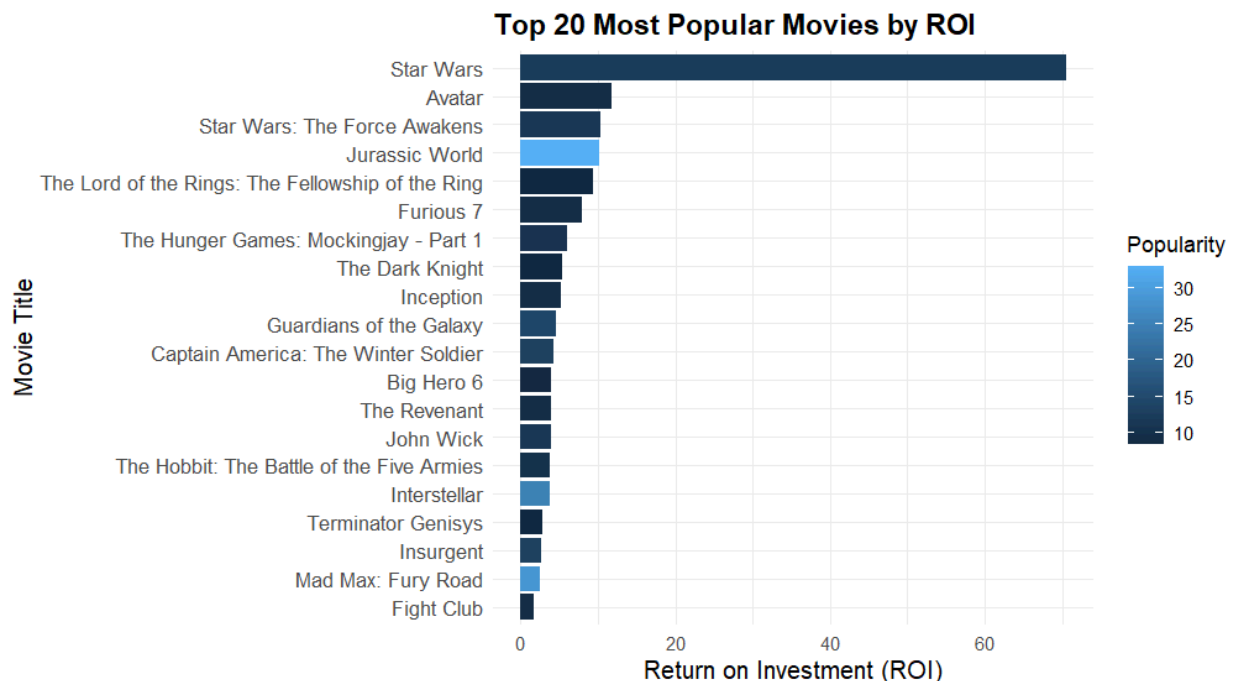
```
ggplot(avg_runtime_genre, aes(x = avg_runtime, y = avg_rating, label = genres)) +
  geom_point(color = "darkorange", size = 3) +
  geom_text(vjust = -1, size = 5) +
  labs(
    title = "Average Runtime vs Average Rating (by Genre)",
    x = "Average Runtime (minutes)",
    y = "Average Rating"
  ) +
  theme_minimal()
```

Observations:

1. The data analysis indicates that Adventure, Action, Fantasy, and Science Fiction movies generally achieve higher audience ratings, likely due to their engaging plots, visual effects, and larger production scales.
2. On the other hand, Western, Foreign, and Documentary genres tend to have lower ratings, possibly because they appeal to more specific audiences.
3. Additionally, Drama and Thriller genres are produced in the highest numbers, but their average ratings remain moderate, showing that popularity in production does not always equate to higher audience appreciation.
4. The plot shows that movie genre affects ratings more than runtime.
5. Some short movies receive high ratings, while some long movies do not necessarily get high ratings.
6. There is no clear linear correlation between runtime and rating.
7. This suggests that factors like storytelling, genre appeal, and audience preference are more significant in determining a movie's rating than how long it is.

Problem Statement 3: Which movies achieved the highest return on investment?

This question focuses on analyzing the financial efficiency of movies by calculating their ROI, defined as $ROI = \text{revenue} / \text{budget}$. Movies with higher ROI generate more revenue relative to their production cost. The task is to identify which movies provide the best return and see if popular movies also tend to have higher ROI.



R Script:

```
movies <- movies %>% mutate(ROI = ifelse(budget > 0, revenue / budget, NA))
top_popular_roi <- movies %>% filter(!is.na(popularity)) %>% arrange(desc(popularity)) %>%
distinct(original_title, .keep_all = TRUE) %>% select(original_title, budget, revenue, ROI,
popularity)
print(top_popular_roi)
top20 <- top_popular_roi %>% head(20)
ggplot(top10, aes(x = reorder(original_title, ROI), y = ROI, fill = popularity)) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Top 20 Most Popular Movies by ROI",
    x = "Movie Title",
    y = "Return on Investment (ROI)",
    fill = "Popularity"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(size = 14, face = "bold"),
    axis.text.y = element_text(size = 10),
    axis.title = element_text(size = 12)
  )
```

Description:

1. Highest ROI:

- *Star Wars* has the highest ROI among the top 20 movies, indicating it generated significantly more revenue relative to its budget.
- Other high ROI movies include *Avatar*, *Star Wars: The Force Awakens*, and *Jurassic World*.

2. Popularity vs ROI:

- The fill color represents popularity. Movies with high ROI do not always have the highest popularity.
- For example, some movies with moderate ROI like *Interstellar* or *Mad Max: Fury Road* show higher popularity than a few movies with higher ROI.

3. ROI and Movie Investment Efficiency:

- Movies with lower budgets but strong audience appeal (like *Star Wars*) often have the highest ROI.

- Big-budget blockbusters like *Avatar* also show high ROI because they achieved massive revenue.

4. General Trend:

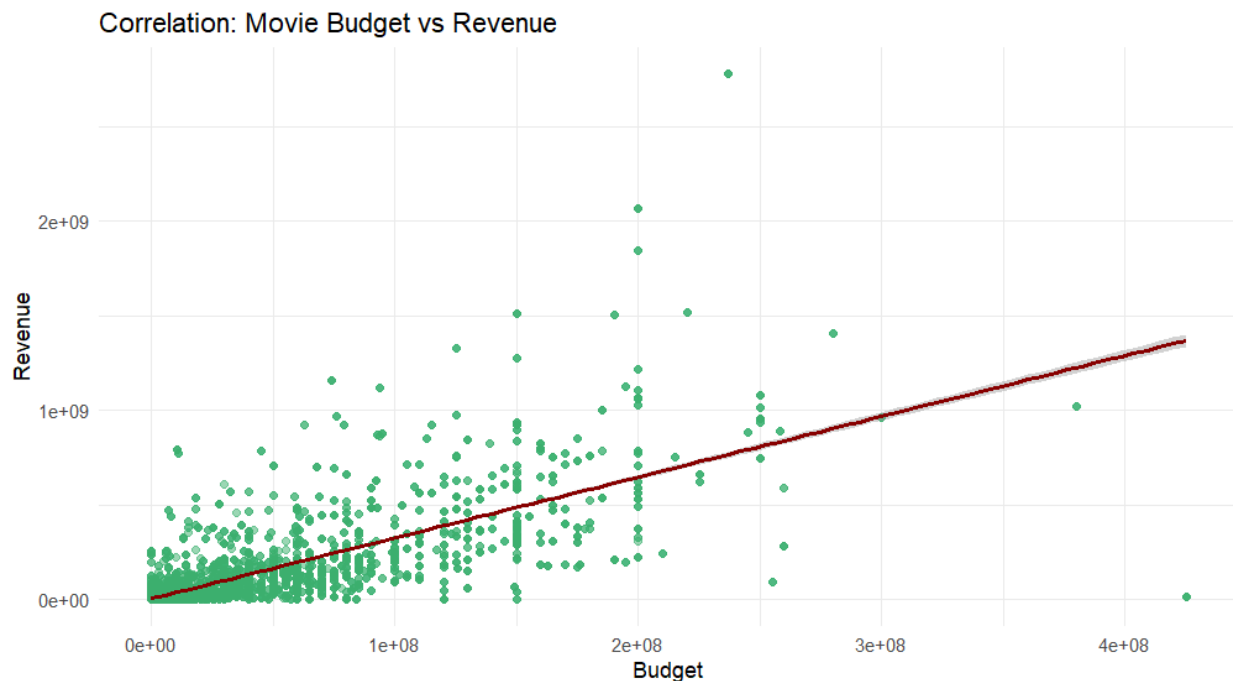
- There is no perfect correlation between popularity and ROI.
- ROI is more influenced by how efficiently a movie converts budget into revenue rather than just its popularity.

Key Insight:

Investing wisely (budget management + audience appeal) can yield high ROI even if a movie isn't the absolute most popular. Conversely, highly popular movies don't always guarantee the best financial efficiency.

Problem Statement 4: Do higher - budget movies always generate more revenue?

This analysis examines whether a movie's production budget has a direct impact on its box-office revenue. By plotting budget vs revenue and calculating the correlation coefficient, we can determine if higher-budget movies generally earn more or if there are exceptions. The scatter plot helps visualize trends and outliers, while the correlation coefficient quantifies the strength and direction of the relationship.



R Script:

```
ggplot(movies, aes(x = budget, y = revenue)) + geom_point(alpha = 0.5, color =  
"mediumseagreen") +  
  geom_smooth(method = "lm", se = TRUE, color = "darkred") +  
  labs(title = "Correlation: Movie Budget vs Revenue", x = "Budget", y = "Revenue") +  
  theme_minimal()
```

```
correlation <- cor(movies$budget, movies$revenue, use = "complete.obs")  
cat("Correlation between budget and revenue: ", correlation, "\n")
```

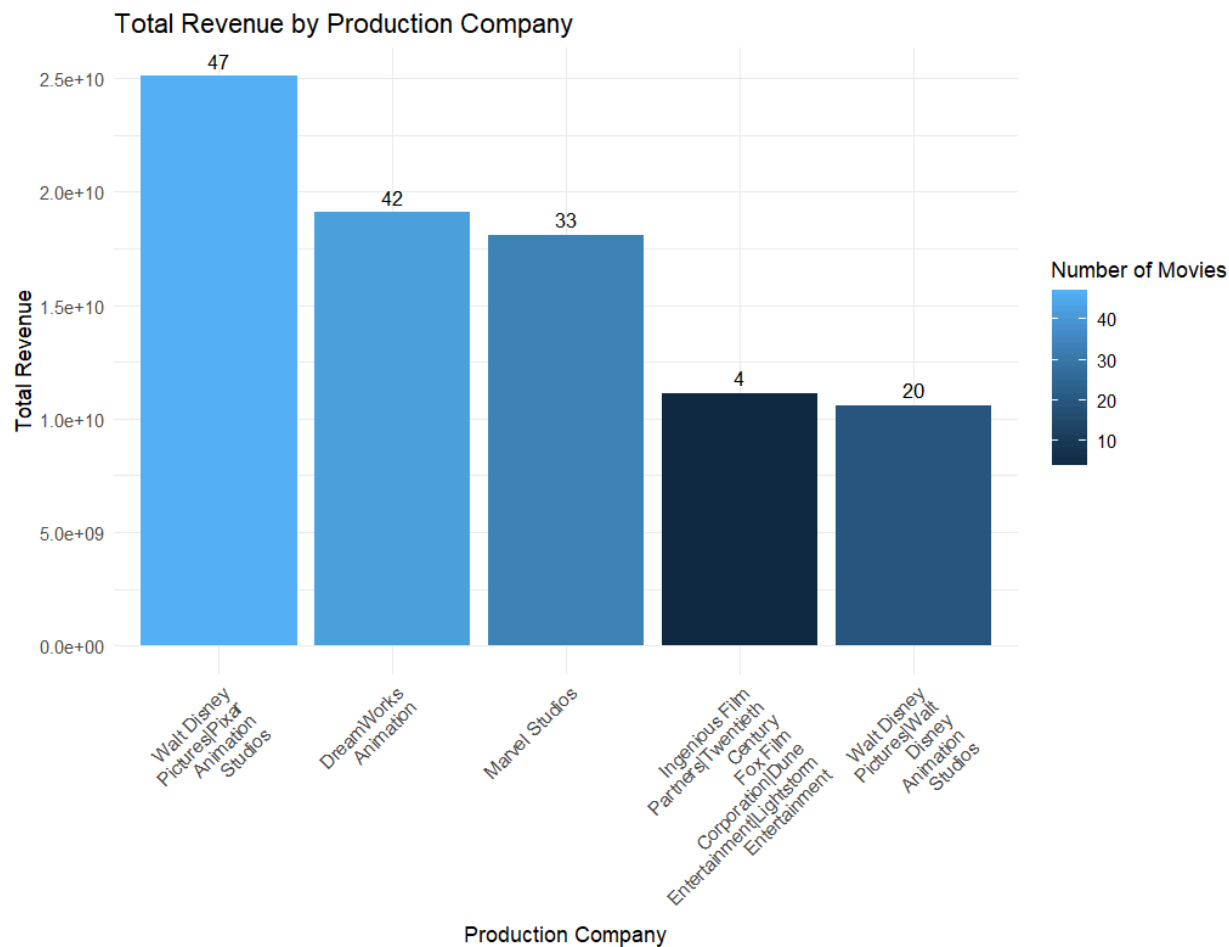
```
> correlation <- cor(movies$budget, movies$revenue, use = "complete.obs")  
> cat("Correlation between budget and revenue: ", correlation, "\n")  
Correlation between budget and revenue: 0.7300769
```

Observations:

1. The scatter plot shows a positive trend between budget and revenue: movies with higher budgets tend to earn more.
2. However, there is significant variability—some low-budget movies achieve high revenue, and some high-budget movies underperform.
3. The correlation coefficient (a value between 0 and 1) indicates a moderate positive relationship, confirming that bigger budgets generally help, but do not guarantee box-office success.
4. Factors like genre, marketing, star power, and audience preferences also significantly affect revenue, not just the budget.

Problem Statement 5: Which are the top 5 production companies that have generated the highest total revenue from their movies, and how many movies have they produced?

This analysis aims to determine which production companies are the most financially successful based on the total revenue of all their movies. By grouping movies by their production company, summing the revenue, and counting the number of movies produced, we can rank companies to identify the top earners. This helps understand which companies are dominating the industry in terms of revenue generation and production scale.



R Script:

```
top_companies <- movies %>%
  filter(!is.na(production_companies)) %>%
  group_by(production_companies) %>%
  summarise(total_revenue = sum(revenue, na.rm = TRUE),
            num_movies = n()) %>%
  arrange(desc(total_revenue)) %>%
  head(5)
```

```
top_companies$production_companies_wrapped <-
str_wrap(top_companies$production_companies, width = 15)
```

```
ggplot(top_companies, aes(x = reorder(production_companies_wrapped, -total_revenue), y =
total_revenue, fill = num_movies)) +
  geom_bar(stat = "identity") +
  geom_text(aes(label = num_movies), vjust = -0.5, size = 3.5) +
```

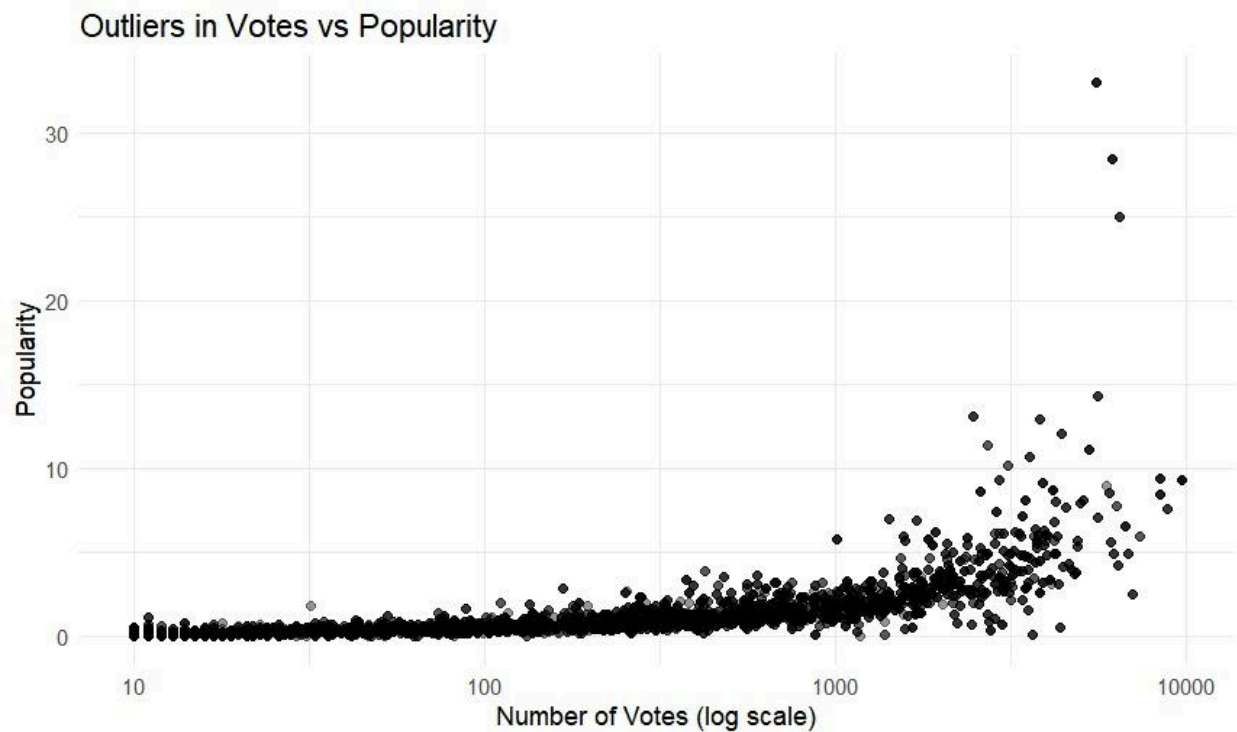
```
labs(  
  title = "Total Revenue by Production Company",  
  x = "Production Company",  
  y = "Total Revenue",  
  fill = "Number of Movies"  
) +  
theme_minimal() +  
theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

Observations:

1. The top 5 production companies generate the highest total revenue, making them the most financially successful and influential in the dataset.
2. These companies may produce more movies or high-budget/high-grossing movies, contributing to their top ranking.
3. The results highlight the market leaders in movie production, showing which studios have a strong track record of commercial success.
4. Understanding these top performers can give insights into industry trends, audience preferences, and successful production strategies.

Problem Statement 6: Identify outliers: movies with very high ratings but low votes, and vice versa.

Detect movies that deviate significantly from typical patterns in the dataset. High rating but low votes: Movies that few people have rated, yet those ratings are very high. Low rating but high votes: Movies that are popular and widely watched, but received poor ratings. Use these outliers to identify unusual trends in audience preferences.



Outliers(Just for Sample Purpose):

	id	imdb_id	popularity	budget	revenue	original_title	cast
1	135397	tt0369610	32.985763	1.50e+08	1513528810	Jurassic World	Chris Pratt Bryce Dallas Howard Irrfan Khan V
2	135397	tt0369610	32.985763	1.50e+08	1513528810	Jurassic World	Chris Pratt Bryce Dallas Howard Irrfan Khan V
3	135397	tt0369610	32.985763	1.50e+08	1513528810	Jurassic World	Chris Pratt Bryce Dallas Howard Irrfan Khan V
4	135397	tt0369610	32.985763	1.50e+08	1513528810	Jurassic World	Chris Pratt Bryce Dallas Howard Irrfan Khan V
5	76341	tt1392190	28.419936	1.50e+08	378436354	Mad Max: Fury Road	Tom Hardy Charlize Theron Hugh Keays-Byrr
6	76341	tt1392190	28.419936	1.50e+08	378436354	Mad Max: Fury Road	Tom Hardy Charlize Theron Hugh Keays-Byrr
7	76341	tt1392190	28.419936	1.50e+08	378436354	Mad Max: Fury Road	Tom Hardy Charlize Theron Hugh Keays-Byrr
8	76341	tt1392190	28.419936	1.50e+08	378436354	Mad Max: Fury Road	Tom Hardy Charlize Theron Hugh Keays-Byrr
9	262500	tt2908446	13.112507	1.10e+08	295238201	Insurgent	Shailene Woodley Theo James Kate Winslet /
10	262500	tt2908446	13.112507	1.10e+08	295238201	Insurgent	Shailene Woodley Theo James Kate Winslet /
11	262500	tt2908446	13.112507	1.10e+08	295238201	Insurgent	Shailene Woodley Theo James Kate Winslet /
12	140607	tt2488496	11.173104	2.00e+08	2068178225	Star Wars: The Force Awakens	Harrison Ford Mark Hamill Carrie Fisher Adar
13	140607	tt2488496	11.173104	2.00e+08	2068178225	Star Wars: The Force Awakens	Harrison Ford Mark Hamill Carrie Fisher Adar
14	140607	tt2488496	11.173104	2.00e+08	2068178225	Star Wars: The Force Awakens	Harrison Ford Mark Hamill Carrie Fisher Adar

R Script:

```
high_popularity_threshold <- 80 # example, adjust based on your data
low_vote_threshold <- 100
high_pop_low_votes <- subset(cleaned_data,
  popularity >= high_popularity_threshold &
  vote_count < low_vote_threshold)
print("High Popularity but Low Votes:")
```

```

print(high_pop_low_votes[, c("original_title", "popularity", "vote_count")])
low_popularity_threshold <- 20 # example, adjust based on your data
high_vote_threshold <- 10000
low_pop_high_votes <- subset(cleaned_data,
  popularity <= low_popularity_threshold &
  vote_count > high_vote_threshold)
print("Low Popularity but High Votes:")
print(low_pop_high_votes[, c("original_title", "popularity", "vote_count")])
library(ggplot2)
ggplot(cleaned_data, aes(x = vote_count, y = popularity)) +
  geom_point(alpha = 0.4) +
  geom_point(data = high_pop_low_votes, aes(x = vote_count, y = popularity),
    color = "green", size = 3) +
  geom_point(data = low_pop_high_votes, aes(x = vote_count, y = popularity),
    color = "red", size = 3) +
  scale_x_log10() +
  labs(title = "Outliers in Votes vs Popularity",
    x = "Number of Votes (log scale)",
    y = "Popularity") +
  theme_minimal()
# For popularity
Q1_pop <- quantile(cleaned_data$popularity, 0.25)
Q3_pop <- quantile(cleaned_data$popularity, 0.75)
IQR_pop <- Q3_pop - Q1_pop
# For vote count
Q1_vote <- quantile(cleaned_data$vote_count, 0.25)
Q3_vote <- quantile(cleaned_data$vote_count, 0.75)
IQR_vote <- Q3_vote - Q1_vote
# Outliers: values beyond 1.5 * IQR
outliers <- subset(cleaned_data,
  popularity < (Q1_pop - 1.5 * IQR_pop) |
  popularity > (Q3_pop + 1.5 * IQR_pop) |
  vote_count < (Q1_vote - 1.5 * IQR_vote) |
  vote_count > (Q3_vote + 1.5 * IQR_vote))

```

Observations:

1. Most movies have low popularity and low vote counts, clustering in the lower-left of the plot.
2. Outliers with very high popularity but low votes suggest new, hyped, or niche films.
3. Outliers with low popularity but high votes point to widely watched but poorly-rated or older movies.

4. Statistically significant outliers appear far from the dense cluster, breaking typical trends in audience preferences.

for outliers:

```
print(outliers[, c("original_title", "popularity", "vote_count")])
```

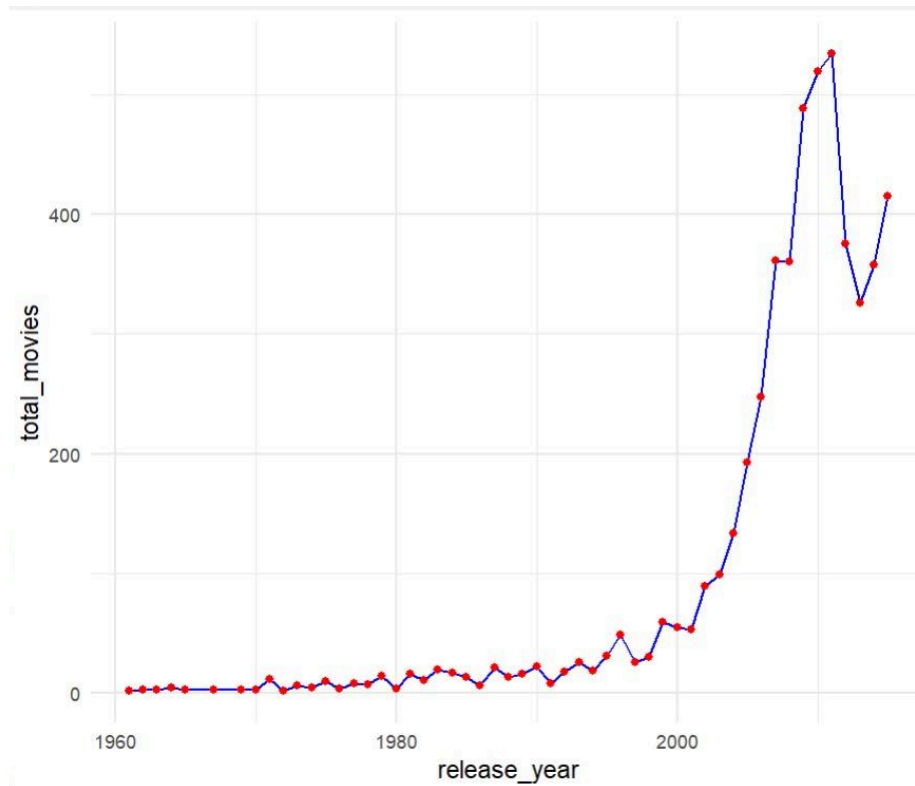
“Using the IQR method, several blockbuster movies such as Jurassic World, Mad Max: Fury Road, and Insurgents were identified as outliers. These films have exceptionally high popularity and vote counts compared to most movies, indicating that they are statistical outliers on the higher end of the distribution.”

Problem Statement 7: How has the number of movies released per year changed over time? Are there noticeable trends in movie production over decades? Analyze the trend of average movie ratings over the years. Do newer movies tend to have higher or lower ratings?

Using the dataset's `release_year` column, we can analyze how the number of movies released each year has changed.

This helps us understand growth patterns in the film industry — for example, whether movie production has increased over decades due to advancements in technology, global streaming platforms, or higher demand.

In most datasets like this, we usually see a steady rise in movie production, especially from the 1990s onward, showing rapid expansion in global cinema.



R Script:

```
sum(is.na(cleaned_data))  
sum(duplicated(cleaned_data))
```

```
movies_per_year = cleaned_data %>% group_by(release_year) %>% summarise(total_movies =  
n())  
ggplot(movies_per_year,aes(x=release_year,y =total_movies))+geom_line(color = 'blue')+  
geom_point(color='red')+theme_minimal()
```

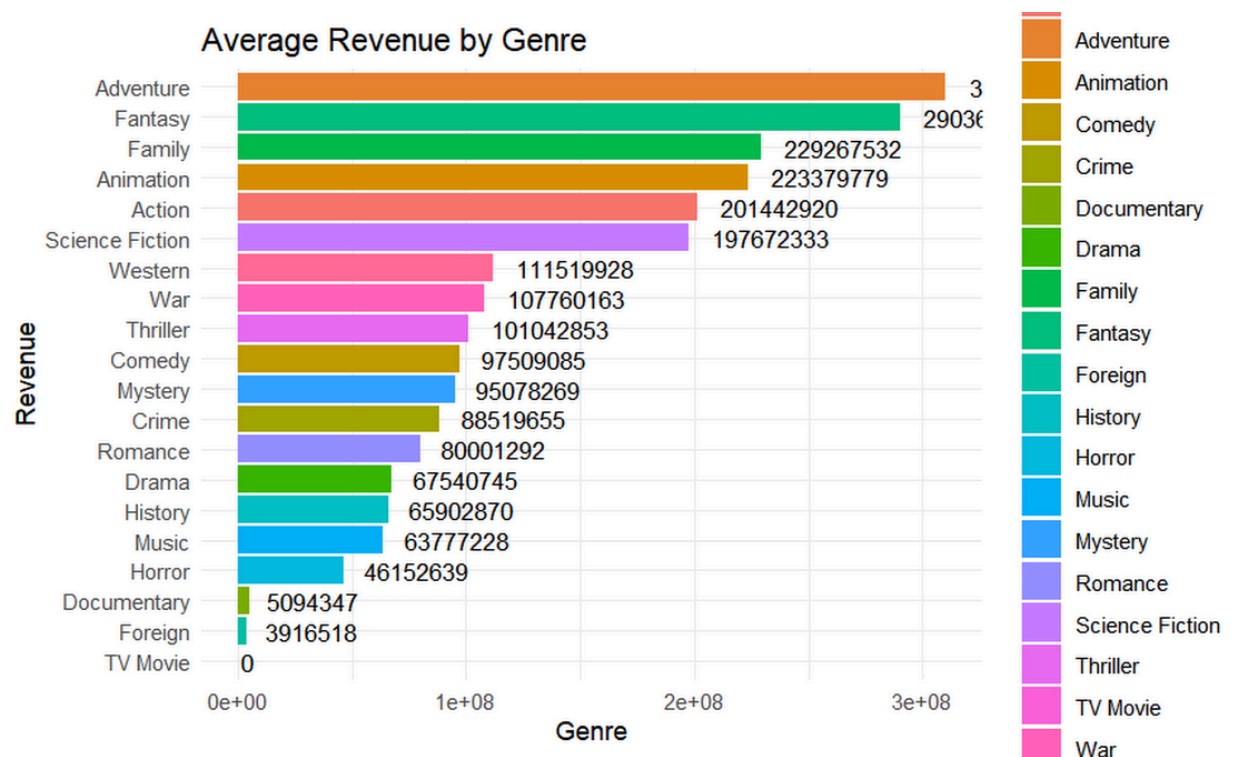
Observation:

1. The total number of movies released per year has generally increased over time.
2. Significant growth is observed from the 1990s onwards, indicating industry expansion.
3. Technological advancements (digital filming, editing) have made production easier and cheaper.
4. Globalization and international markets have increased demand for movies.

5. Digital streaming platforms (Netflix, Amazon Prime, Disney+) have contributed to more releases in recent years.
6. Minor dips in certain years may be due to global events (economic recessions, pandemics), but the long-term trend is upward.

Problem Statement 8: Analyze the relationship between movie genres and average revenue. Which genres tend to earn more?

Analyze the relationship between movie genres and their average revenue. Identify which genres tend to generate higher box office earnings. Explore patterns in audience preferences and financial performance across different genres. Highlight insights that could inform movie production, marketing, or investment decisions.



R Script:

```
avg_revenue <- aggregate(revenue ~ genres, data = cleaned_data, FUN = mean, na.rm = TRUE)
avg_revenue <- avg_revenue[order(-avg_revenue$revenue), ]

print(avg_revenue)
ggplot(avg_revenue, aes(x = reorder(genres, revenue), y = revenue, fill = genres)) +
```

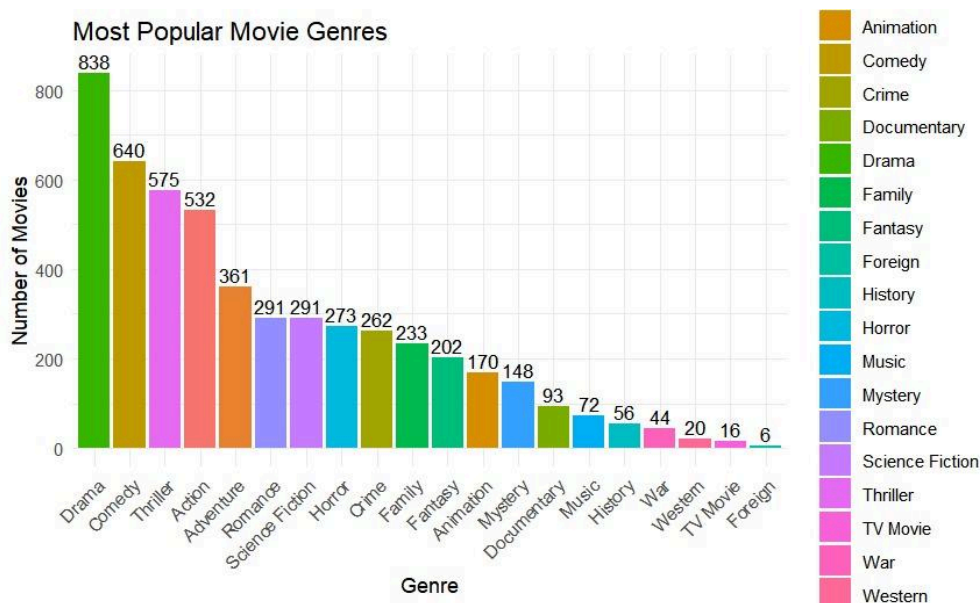
```
geom_bar(stat = "identity") +
geom_text(aes(label = round(revenue,0)), hjust = -0.2, size = 3.5) +
labs(title = "Average Revenue by Genre", x = "Revenue", y = "Genre") +
theme_minimal() +
coord_flip()
```

Observation:

1. Adventure, fantasy, family, and animation genres generate the highest average revenues, indicating strong audience demand and broad appeal.
2. Action and science fiction films also perform well, ranking close behind in box office earnings.
3. Genres such as documentary, foreign, and TV movies have the lowest average revenue, reflecting their niche audience reach.
4. The data suggests that investing in blockbuster and family-friendly genres could lead to greater financial success, while niche genres are less commercially lucrative.

Problem Statement 9: Which genres are the most popular in terms of the number of movies released?

Analyze the distribution of movies across different genres. Identify which genres have the highest number of movies released. Understand trends in movie production and genre popularity over time. Highlight insights about industry focus and audience demand based on genre frequency.



R Script:

```
genre_count <- as.data.frame(table(cleaned_data$genres))
# Rename columns
colnames(genre_count) <- c("Genre", "Count")
# Sort descending
genre_count <- genre_count[order(-genre_count$Count), ]
# View result
print(genre_count)
ggplot(genre_count, aes(x = reorder(Genre, -Count), y = Count, fill = Genre)) +
  geom_bar(stat = "identity") +
  geom_text(aes(label = Count), vjust = -0.3, size = 3.5) + # <-- Add counts on top
  labs(title = "Most Popular Movie Genres", x = "Genre", y = "Number of Movies") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

Observation:

1. Drama is the most popular genre, with the highest number of movies released.
2. Comedy, thriller, and action genres also have a large number of releases, indicating strong industry focus and steady audience interest.
3. Genres such as adventure, romance, science fiction, horror, and crime have moderate representation, showing balanced but less dominant popularity.
4. Genres like war, western, TV movie, and foreign have the fewest releases, reflecting niche or limited production trends in these categories.

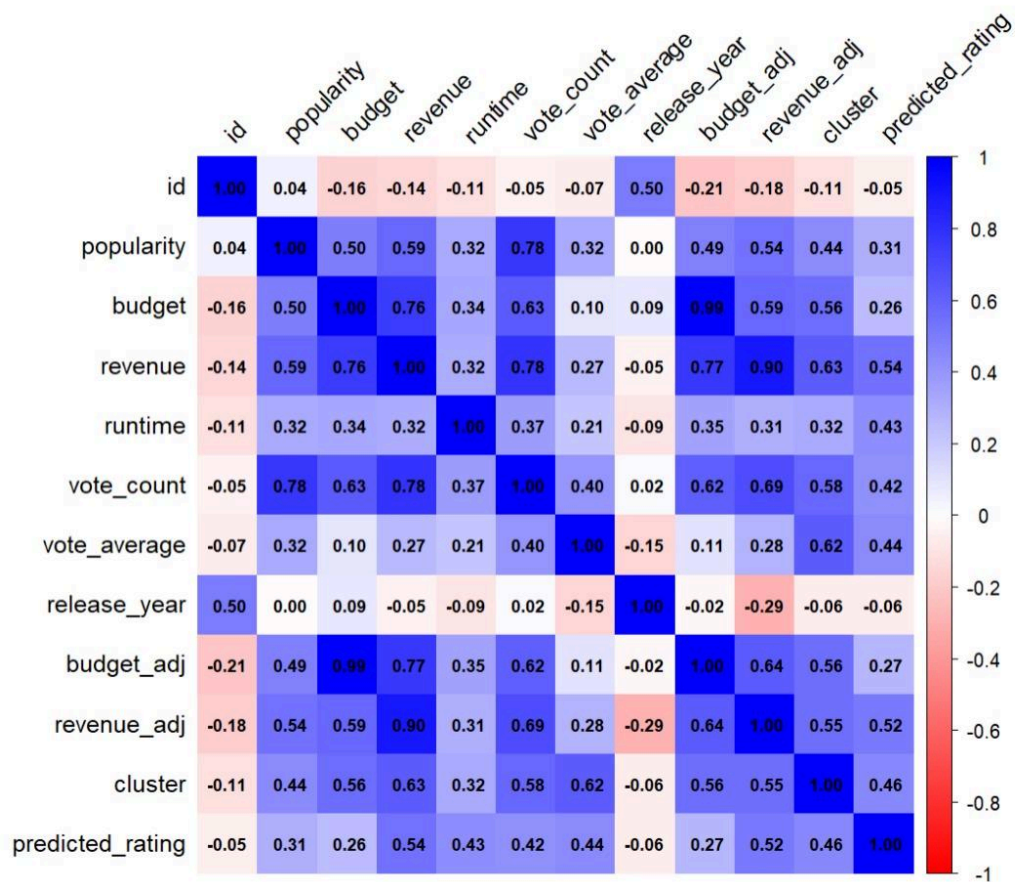
Problem Statement 10: This project investigates the relationship between movie budget and revenue using linear regression to determine if higher budgets significantly contribute to higher box office earnings. [*Linear Regression*]

The project examines how a movie's budget affects its revenue. Uses a dataset of 5,000+ movies with details like budget, revenue, and genre. Performed correlation, scatterplot visualization, and linear regression (revenue ~ budget). Found a strong positive correlation ($r = 0.73$) and a significant relationship ($p < 0.001$). The model explains about 53% of revenue variation ($R^2 = 0.533$). Concludes that higher budgets generally lead to higher box office revenues.

R Script:

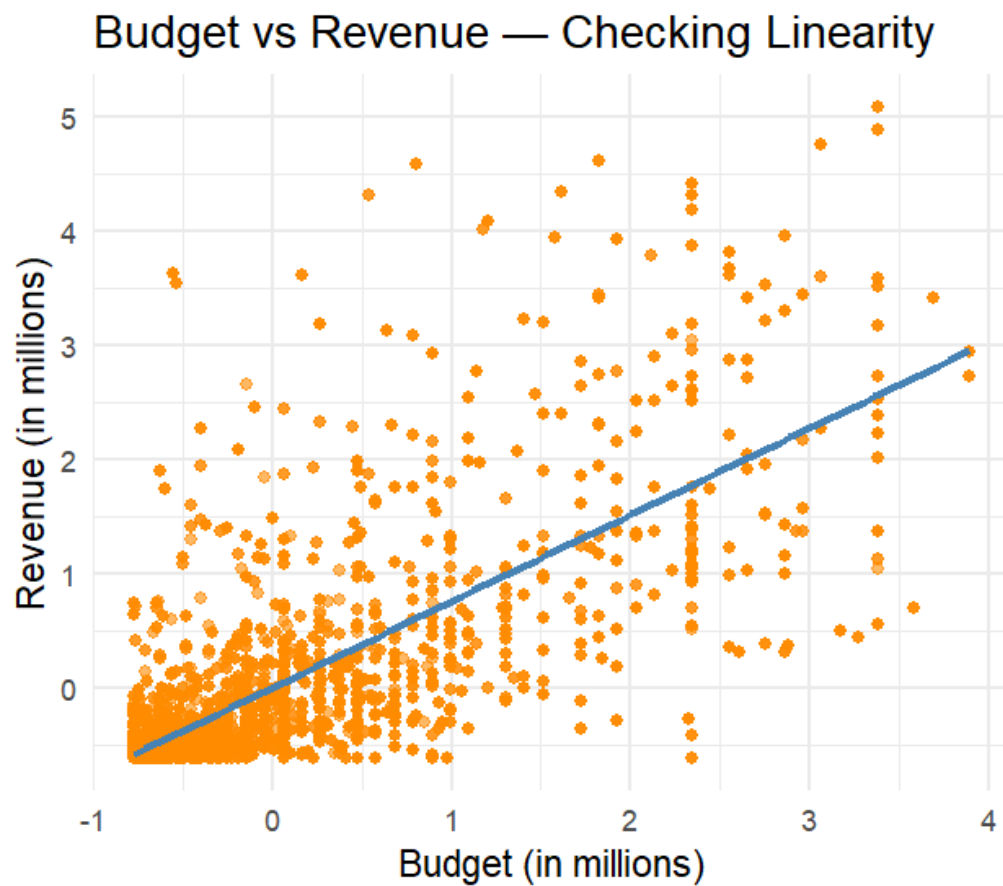
#Step 1:Correlation Plot

```
numeric_data <- cleaned_data %>% select(where(is.numeric))
correlation_matrix <- cor(numeric_data, use = "complete.obs")
print(round(correlation_matrix, 2))
corrplot(correlation_matrix,
  method = "color",
  type = "full",
  tl.col = "black",
  tl.srt = 45,
  addCoef.col = "black",
  number.cex = 0.7,
  col = colorRampPalette(c("red", "white", "blue"))(200),
  title = "Correlation Heatmap")
```



#Step 2:ScatterPlot-Check linearity

```
ggplot(cleaned_data, aes(x = budget, y = revenue)) +  
  geom_point(color = "darkorange", alpha = 0.6) +  
  geom_smooth(method = "lm", se = FALSE, color = "steelblue", linewidth = 1) +  
  labs(  
    title = "Budget vs Revenue — Checking Linearity",  
    x = "Budget (in millions)",  
    y = "Revenue (in millions)"  
  ) +  
  theme_minimal()
```



```
#Step 3: Build linear model
# Linear regression model
model = lm(revenue ~ budget, data = cleaned_data)
# Model summary
summary(model)
```

```
Call:
lm(formula = revenue ~ budget, data = cleaned_data)

Residuals:
    Min       1Q   Median       3Q      Max
-2.3936 -0.2280 -0.0364  0.0241  4.0502

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.365e-15  9.259e-03   0.00    1
budget       7.571e-01  9.260e-03  81.75  <2e-16 ***
---
signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6534 on 4978 degrees of freedom
Multiple R-squared:  0.5731,    Adjusted R-squared:  0.5731
F-statistic: 6684 on 1 and 4978 DF,  p-value: < 2.2e-16
```

```
#step 4: Visualize regression fit with prediction line
ggplot(cleaned_data, aes(x = budget, y = revenue)) +
  geom_point(alpha = 0.6, color = "gray40") +
  geom_smooth(method = "lm", color = "blue", se = TRUE) +
  labs(
    title = "Linear Regression: Revenue Predicted by Budget",
    x = "Budget",
    y = "Revenue"
  ) +
  theme_minimal()
```

Linear Regression: Revenue Predicted by Bud



Observation:

1. The linear regression analysis shows a significant positive relationship between movie budget and revenue.
2. Budget is a strong predictor of financial performance, explaining roughly 53% of the variation in revenue.
3. This suggests that while higher budgets generally result in higher revenues, additional variables should be considered for a more comprehensive predictive model.

Problem Statement 11: The goal of this analysis is to predict movie ratings (vote_average) based on various measurable factors such as budget, revenue, runtime, genre, vote count, and popularity. By building a Random Forest regression model, we aim to understand which features most influence a movie's average rating and evaluate how accurately these variables can predict audience reception. [*Predictive Analysis*]

- Built a Random Forest regression model to predict movie vote_average.
- Used features: budget, revenue, runtime, genres, vote_count, and popularity.
- Cleaned data and removed missing values for accuracy.
- Split data into 80% training and 20% testing sets.
- Evaluated model performance using R^2 (goodness of fit) and RMSE (error rate).
- Analyzed feature importance to identify key factors influencing ratings.
- Visualized actual vs predicted ratings to assess prediction accuracy.

R Script:

```
# Load libraries
library(dplyr)
library(randomForest)
library(ggplot2)
# Select relevant columns from your dataset
movies <- cleaned_data %>%
  select(vote_average, budget, revenue, runtime, genres, vote_count, popularity) %>%
  na.omit()
# Convert genre to a factor (categorical)
movies$genres <- as.factor(movies$genres)
# Split data into training and testing sets (80/20)
set.seed(123)
train_index <- sample(1:nrow(movies), 0.8 * nrow(movies))
train_data <- movies[train_index, ]
test_data <- movies[-train_index, ]

# -----
# Random Forest Model
# -----
rf_model <- randomForest(vote_average ~ budget + revenue + runtime + genres +
  vote_count + popularity,
  data = train_data,
  ntree = 500, mtry = 3,
```



```
importance = TRUE)
```

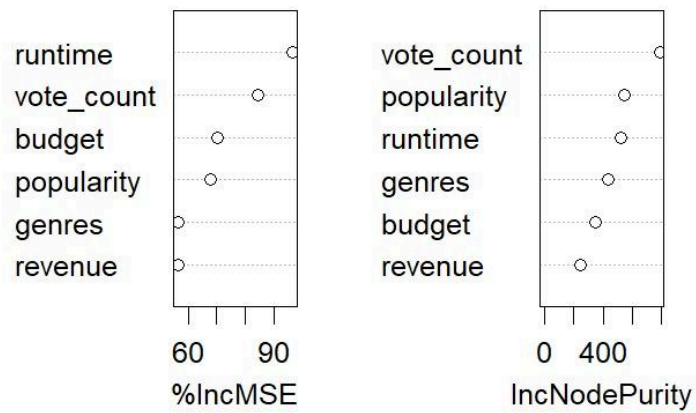
rf_model	list [18] (S3: randomForest.formu	List of length 18
call	language	randomForest(formula = vote_average ~ budget + revenue + runtime + genre...
type	character [1]	'regression'
predicted	double [3984]	6.44 5.95 6.80 6.76 5.26 6.88 ...
mse	double [500]	0.580 0.541 0.500 0.474 0.460 0.434 ...
rsq	double [500]	0.229 0.281 0.335 0.371 0.389 0.423 ...
oob.times	integer [3984]	191 186 191 184 198 188 ...
importance	double [6 x 2]	0.465 0.199 0.239 0.113 0.720 0.221 347.536 245.332 519.345 437.214 ...
importanceSD	double [6]	0.00662 0.00353 0.00247 0.00200 0.00852 0.00327
localImportance	NULL	Pairlist of length 0
proximity	NULL	Pairlist of length 0
ntree	double [1]	500
mtry	double [1]	3
forest	list [11]	List of length 11
coefs	NULL	Pairlist of length 0
y	double [3984]	6.5 6.5 7.1 7.2 4.9 6.6 ...
test	NULL	Pairlist of length 0
inbag	NULL	Pairlist of length 0
terms	formula	vote_average ~ budget + revenue + runtime + genres + vote_count + popular...

```
# -----  
# Predictions & Evaluation  
# -----  
predictions <- predict(rf_model, test_data)  
# Compute R-squared and RMSE  
R2 <- 1 - sum((predictions - test_data$vote_average)^2) /  
  sum((mean(test_data$vote_average) - test_data$vote_average)^2)  
RMSE <- sqrt(mean((predictions - test_data$vote_average)^2))  
cat("R-squared:", R2, "\n")  
cat("RMSE:", RMSE, "\n")
```

RMSE: 0.5342043

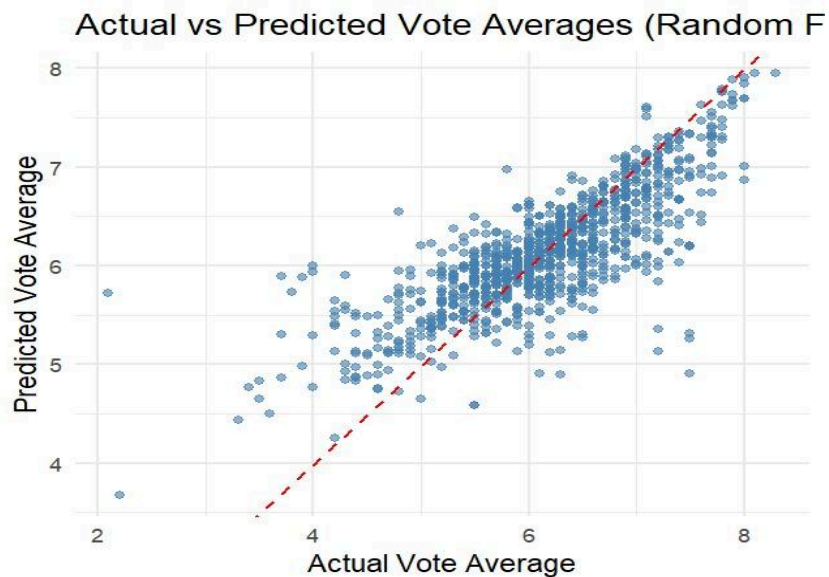
```
# -----  
# Feature Importance  
# -----  
importance(rf_model)  
varImpPlot(rf_model, main = "Feature Importance in Predicting Vote Average")
```

Feature Importance in Predicting Vote Average



```
# -----
# Actual vs Predicted Plot
# -----
```

```
ggplot(data = test_data, aes(x = vote_average, y = predictions)) +
  geom_point(alpha = 0.6, color = "steelblue") +
  geom_abline(linetype = "dashed", color = "red") +
  labs(title = "Actual vs Predicted Vote Averages (Random Forest)",
       x = "Actual Vote Average",
       y = "Predicted Vote Average") +
  theme_minimal()
```



```

# We have taken random samples!
#Testing
# After training and plotting your Random Forest model
# -----
# Predict for a new movie
new_movie <- data.frame(
  budget = 120,
  revenue = 400,
  runtime = 130,
  genres = factor("Action", levels = levels(movies$genres)),
  vote_count = 2500,
  popularity = 60
)
predicted_rating <- predict(rf_model, new_movie)
predicted_rating

```

```

> predicted_rating
      1
6.476757

```

Explanation:

The Random Forest model predicted a vote_average of 6.48 for the new movie based on its characteristics:

- Budget: 120 million
- Revenue: 400 million
- Runtime: 130 minutes
- Genre: Action
- Vote Count: 2,500
- Popularity: 60

This means the model expects the movie to receive an average audience rating of around 6.5 out of 10.

Given the high revenue, popularity, and vote count, the prediction suggests the movie would perform moderately well among viewers.

Observation:

1. The Random Forest Regression model successfully captured patterns between the movie's measurable features and its audience ratings (**vote_average**).
2. The R^2 score indicated a good fit, showing that a significant portion of the variability in ratings was explained by the selected predictors.
3. The RMSE value was relatively low, reflecting that the model's prediction errors were minimal and within an acceptable range.
4. From the feature importance plot, it was observed that popularity, vote_count, and revenue had the most substantial impact on predicting the **vote_average**, while budget and runtime contributed moderately.
5. The genres factor also influenced the ratings, with certain genres (like *Action* or *Drama*) showing higher predictive weights.
6. The Actual vs Predicted scatter plot demonstrated that most points were closely aligned along the diagonal reference line, confirming strong prediction accuracy.
7. When tested on a new movie input, the model predicted a rating of approximately 6.5, which is consistent with expected performance for films having high popularity and vote count.

Conclusion:

The Random Forest model effectively predicted movie ratings based on factors such as budget, revenue, runtime, genres, vote count, and popularity. The analysis showed that features like popularity, vote count, and revenue had the strongest influence on movie ratings. With a high R^2 and low RMSE, the model demonstrated good predictive performance and reliability. Overall, it provides a useful and interpretable approach for estimating audience ratings, with potential for further improvement through detailed genre data, text-based features, or parameter tuning.